Audio-Visual Navigation with Anti-Backtracking

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Abstract. Embodied navigation, which involves robotic agents exploring an unknown environment to reach target locations with egocentric observation, is a complex problem in the field of embodied AI. Audio-visual navigation extends this concept by equipping agents with both visual and auditory sensors. Recent studies have explored the audio-visual navigation task, exploring its potential and intricacies. Current methodologies, despite their achievements, often fail to fully utilize the capabilities of these sensory modalities, leading to sub-optimal designs and inefficiencies. In addition, the prevalent backtracking problem in navigation tasks often leads to redundant movements by robots. To address these challenges, we introduce the Audio-Visual Guided Navigation (AVGN) model. At its core, AVGN champions the fusion of visual and auditory data through sophisticated modality fusion layers. Our model represents a significant advancement by employing transformers for visual encoding and deploying a novel Dual Stage Feature Integration (DSFI) to decode latent interrelationships between the visual and audio realms. With AVGN, the backtracking issue is mitigated, as the acoustic map synergizes more effectively with visual features for informed action decisions. Furthermore, targeting the backtracking problem, we propose a unique set of reward structures to guide and refine the actions of the agent. Benchmark evaluations on Replica and Matterport3D datasets validate our claims, and AVGN notably surpasses existing methodologies in audio-visual navigation tasks.

Keywords: Visual Navigation · Reinforcement Learning · Audio Visual Navigation.

1 Introduction

Embodied navigation [1]–[3] is a critical area of research in the field of embodied intelligence [4], [5], where robotic agents navigate unknown environments to reach their target destinations using egocentric observations. In addition to visual observations, hearing is an essential sense as it provides both temporal and spatial information, allowing visually impaired subjects to navigate efficiently. To get a better understanding of physical space and localizing sound-emitting targets, Chen et al. [6] introduced the concept and standard of the audio-visual navigation task. Concurrently, research [6]–[11] have been proposed for the audio-visual navigation task. Among recent audio-visual navigation research, LLA [8]

presents a traditional phase-based navigation approach, leveraging a topological graph to streamline path planning. SoundSpaces [6] distinguishes itself as a trailblazing end-to-end solution, uniquely foregoing reliance on topological graphs and specific auditory meta-data such as sound source categories (e.g., telephone, doorbell, alarm). AV-WaN [11] utilizes waypoints (depicted via a topological graph) to enhance long-distance navigation efficacy. SAVi [10] addresses environments with sporadic sound emissions by incorporating auditory meta-data.

However, a significant limitation of these methodologies lies in their approach to fusing visual and auditory data. Their strategies for modalities integration often use concatenation. This simplistic merging often loses the details that each type of sensor can provide, potentially missing opportunities for deeper insights and enhanced navigation cues. Furthermore, existing methods suffer from the pervasive backtracking problem. This challenge manifests itself in multiple ways: it not only introduces redundancy trajectories, inflating the navigation time, but it can also lead to agents being sent astray. Such misdirection can be particularly detrimental in complex or dynamic environments where efficient navigation is paramount. Agents often find themselves retracing their steps or oscillating between points, increasing the risk of navigational failure.

To enhance the effectiveness of information utilization and reduce the backtracking issue, our proposed methodology aims to effectively integrate both audio and visual cues and examine their interrelationships. We have designed a model that comprises two dedicated encoders, tailored for visual and audio inputs, complemented by an advanced suite of fusion techniques on different levels of features. In contrast to conventional Convolutional Neural Networks (CNNs), vision transformers [12], [13] demonstrate superior capability in capturing intricate spatial relationships and global context from visual data. Leveraging the prowess of transformers, our model becomes adept at extracting and assimilating pivotal features from visual streams. To further accentuate this synergy, we introduce a multi-stage fusion strategy termed Dual Stage Feature Integration (DSFI). This approach allows our model to discern and represent the nuanced connections between visual scenes and concurrent audio cues. The DSFI procedure is bifurcated into two fusion strata. The Early Fusion integrates early visual and audio features, offering a foundational context and immediate sensory correlations. On the other hand, the Acoustic Fusion branch refined visual and auditory features in a more context-aware manner. By introducing DSFI into our architecture, our model can seamlessly blend insights from both sensory channels, providing a holistic and enriched understanding for the navigating agent. Since the acoustic map contains information about the direction of the sound source, which is also the navigation target, the backtracking issue is alleviated. Moreover, in addressing the common backtracking issue inherent in many navigation algorithms, we introduce the directional continuity reward and the historical trajectory reward to constrain the agent's actions. The directional continuity reward penalizes abrupt changes in the agent's direction of movement. The historical trajectory reward imposes penalties if the agent revisits a location to which it has already been on its trajectory.

In conclusion, our contributions can be summarized as three-fold: **(i)** We introduced a novel audio-visual navigation method, AVGN, which employs a robust fusion strategy, DSFI, adeptly integrating visual and audio features to optimize action decisions; **(ii)** We have innovatively devised a triad of reward mechanisms, specifically targeting and mitigating the persistent backtracking challenges inherent to navigation tasks; **(iii)** Benchmarking on the SoundSpaces platform, our model has demonstrated superior performance over existing methods on Matterport3D and Replica datasets.

2 Related Work

2.1 Vision-Based Navigation

The significance of vision in cognitive mapping for human navigation has been extensively investigated in early research [14], [15]. Similarly, recent AI agents process egocentric visual input for navigation purposes [14], [16]–[21]. NTS [16] presents a method for image-goal navigation in uncharted environments by leveraging topological space representations that intricately merge semantics with proximate geometry, hinged on nodes and their relationships. Meanwhile, LB-WayPtNav [14] offers an inventive blend of model-based control and learning-oriented perception for robot navigation, delivering consistent and efficient trajectories in uncharted terrains. Bansal et al. [17] introduces a technique that synergizes model-based control with learning-driven perception to navigate robots, resulting in reliable and efficient trajectories in unfamiliar environments, even with minimal 3D environmental mapping and lower frame rates, showcasing effective sim-to-real generalization.

Vision-based navigation is frequently augmented with textual input in the form of dialogs. Pioneering studies [22]–[24] have investigated the feasibility of vision-dialog navigation. Specifically, Jesse et al. [22] curate a dataset tailored for Cooperative Vision-and-Dialog Navigation set within photorealistic residential environments, and introduce a task where agents rely on human dialog history for navigation. Their insights reveal that a richer dialog history boosts navigation outcomes. VLN-BERT [23] refines the BERT [25] architecture by incorporating a time-sensitive recurrent function for the vision-and-language navigation challenge, showcasing adaptability over multiple transformer models and adeptness in managing both navigation and expression referral tasks. Furthermore, Hao et al. [24] propose a pre-training and fine-tuning strategy for vision-language navigation tasks, which leverages a plethora of image-text-action datasets in a self-supervised manner to augment performance in established vision-language navigation setups.

Although vision-based and vision-language navigation has achieved considerable success, audiovisual navigation is emerging as a novel challenge in the domain.

2.2 Audio-Visual Navigation

Audio-Visual Navigation pertains to tasks where agents process visual and auditory cues. Historically, most audio-visual navigation research has been confined to single-sound environments. Notable examples include LLA (Look, Listen, and Act) [8], SoundSpaces [6], AV-WaN (Audio-Visual Waypoint Navigation) [11], and SAVi (Semantic Audio-Visual Navigation) [10]. LLA [8] offers a conventional phase-based navigation method, relying on a topological graph to facilitate optimal path planning. SoundSpaces [6], a pioneering end-to-end method, stands out by not depending on any topological graph or specific auditory meta-information, such as the category of the sound source (e.g., telephone, doorbell, alarm). AV-WaN [11] adopts waypoints (represented through a topological graph) to optimize navigation over longer distances. SAVi [10] considers scenarios with intermittent sound emissions, enriching their model with auditory meta-information. For more acoustically intricate settings, where the target sound blends with multiple ambient noises, SAAVN (Sound Adversarial Audio-Visual Navigation) [9] offers an end-to-end framework.

Although significant strides have been made in audiovisual navigation, the audio and visual modalities are recognized as underutilized, a gap that we aim to bridge in our proposed methods.

3 The Proposed Method

3.1 Method Overview

We frame the navigation challenge as a reinforcement learning problem, where a robot, driven by an audio cue, learns an optimal policy to swiftly navigate to a target within an unfamiliar environment. We focus on two tasks: AudioGoal Navigation and AudioPointGoal Navigation. In the AudioGoal scenario, the agent receives both audio and visual inputs. AudioPointGoal is an audio extension of the PointGoal task [26]–[28], where the agent navigates with an egocentric view and audio input, and its displacement to the target. The action space of the agent comprises {MoveForward, TurnLeft, TurnRight, Stop}. Following [6], the sensory inputs include binaural sound simulated by room impulse response (RIR) [29], egocentric RGB images, and the displacement vector to the goal (exclusive in the AudioPointGoal setting). We introduce our solution as AVGN, which stands for Audio-Visual Guided Navigation. The architecture of AVGN consists of four key components, as illustrated in Fig. 1. Specifically, AVGN processes given egocentric vision and audio inputs by encoding them into distinct features. Visual data is encoded using transformers, while audio data is processed with convolutional neural networks (CNNs). These separate features are then fused through the Dual Stage Feature Integration (DSFI), resulting in a comprehensive audio-visual embedding. This fused embedding is subsequently transformed into a temporally-sensitive state representation by a Gated Recurrent Unit (GRU). Finally, an actor-critic framework is applied to predict, evaluate, and refine subsequent actions. The robot agent iteratively undergoes this

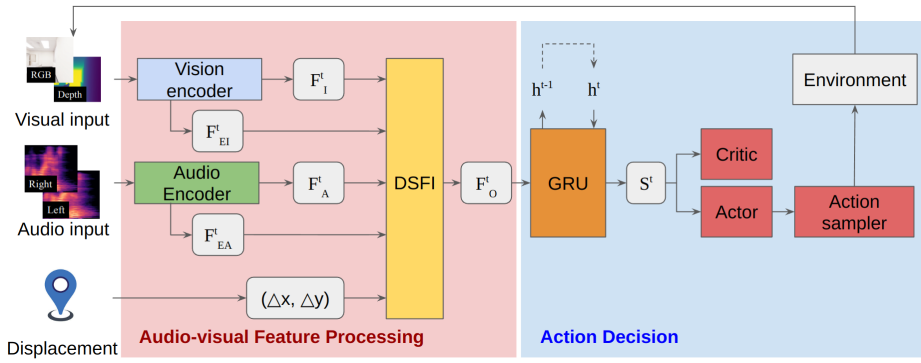

Fig. 1: **The AVGN structure.** The visual and audio inputs are initially processed by their respective encoders. Each type of input is encoded to generate the corresponding features, which include both primary and early-stage characteristics. These features, along with a displacement vector, are then input into the Dual Stage Feature Integration (DSFI) to derive an observation feature. Subsequently, this observation feature is fed into a Gated Recurrent Unit (GRU) to capture the state feature at step t. Finally, this state feature is used by an actor-critic algorithm to determine the appropriate action.

sequence until it successfully locates the target. Each of these components will be elaborated upon in the subsequent sections.

3.2 Audio-Visual Model

At t-th step, the agent receives a visual input, denoted as I^t, and an audio input, represented by A^t. The visual input I^t undergoes processing via a Swin transformer [13], resulting in the derivation of an early image feature, $\mathbf{F}^t_{EI}$, and an image feature, $\mathbf{F}^t_I$. Similarly, the audio input A_t is processed through a Convolutional Neural Network (CNN) to obtain an early audio feature, $\mathbf{F}^t_{EA}$, and an audio feature, $\mathbf{F}^t_A$. Additionally, the GPS sensor of the agent collects the displacement input, Δ^t.

To gauge the real-time contribution of each modality, factoring in the fluctuating contexts, we have devised a trainable audio-visual fusion mechanism, termed Dual Stage Feature Integration (DSFI). This mechanism is adept at converting the encoded features into a consolidated embedding vector, $\mathbf{F}^t_O$:

$$\mathbf{F}^t_O = DSFI(\mathbf{F}^t_I, \mathbf{F}^t_A, \mathbf{F}^t_{EI}, \mathbf{F}^t_{EA}, \Delta^t) \tag{1}$$

The detailed structure of DSFI is illustrated in Fig. 2. The early fusion branch is shown on the left, and we pay special attention to the early visual feature $\mathbf{F}^t_{EI}$ with the early audio feature $\mathbf{F}^t_{EA}$ as follows. For the early audio feature $\mathbf{F}^t_{EA}$, we apply the max pooling and average pooling along the channels. We obtain max pool $\mathbf{F}_{max} \in \mathbb{R}^{1\times W\times H}$ and the average pool $\mathbf{F}_{avg} \in \mathbb{R}^{1\times W\times H}$, respectively. Then

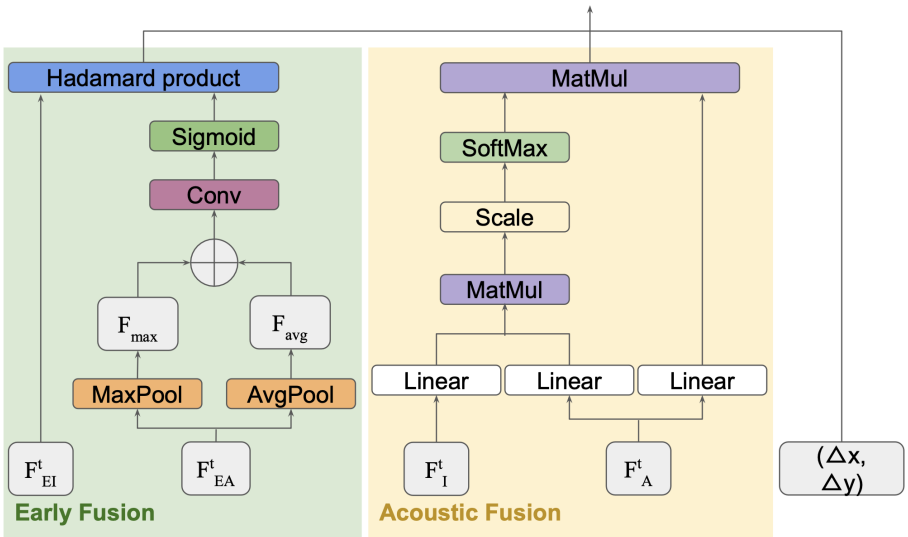

Fig. 2: **The structure of DSFI.** The early audio feature $\mathbf{F}^t_{EA}$ is first enhanced with a spatial attention mechanism and then combined with the early vision features $\mathbf{F}^t_{EI}$ using the Hadamard product. For the image and audio features, we apply cross-attention to process their interaction. The resulting output is then concatenated with the early fused features and the displacement vector to form the observation feature.

we concatenate the obtained features and process a 1×1 convolutional layer followed by a sigmoid function, to obtain a weighted early audio feature. Then we proceed with a Hadamard product between the weighted early audio feature and the early visual feature to obtain an early fusion feature. The procedure above can be concluded as follows:

$$\mathbf{F}^t_E = sigmoid(Conv(\mathbf{F}_{max} \oplus \mathbf{F}_{avg})) \odot \mathbf{F}^t_{EI} \tag{2}$$

The other branch is the acoustic fusion branch, which we process with visual feature $\mathbf{F}^t_I$ and audio feature $\mathbf{F}^t_A$. We apply the function as follows:

$$\mathbf{F}^t = softmax(\frac{\mathbf{W}_1\mathbf{F}^t_I(\mathbf{W}_2\mathbf{F}^t_A)^T}{\sqrt{d}})\mathbf{W}_3\mathbf{F}^t_A \tag{3}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are both weight matrices to be optimized; d is a scalar factor with a value of 256; and F^t is the resulting fused embedding vector. We use the right branch to extract useful information from the audio feature $\mathbf{F}^t_A$ and fuse with the visual feature $\mathbf{F}^t_I$. In this way, the audio information, which delivers the sound source orientation, is combined with the visual feature and the agent can make better decisions accordingly.

The primary advantage of the DSFI design lies in both the acoustic fusion branch and the early fusion branch. Since the acoustic reveals where the sound

comes from, it is important to transfer the audio information into visual features. By fusing the visual feature $\mathbf{F}_I^t$ and the audio feature $\mathbf{F}_A^t$, it ensures a robust correlation between the two, efficiently extracting valuable information from the audio data and enhancing the visual feature in the process. Importantly, the fusion of early visual and audio features delves into finer details of the environment, capturing nuanced environmental cues. Such multi-level feature fusion greatly aids in making more informed decisions regarding actions.

With the fused early feature $\mathbf{F}_E^t$, the fused feature $\mathbf{F}^t$, and the displacement vector Δ, the DSFI concatenates them together to obtain the feature of current observation at step t:

$$\mathbf{F}_O^t = \mathbf{F}_E^t \oplus \mathbf{F}^t \oplus \Delta. \tag{4}$$

We employ a bidirectional GRU [30] with a 512-dimensional hidden layer to enhance a series of observation features, namely $\mathbf{F}_O^{t_0}$ through $\mathbf{F}_O^{t_n}$, culminating in a temporally-aware state representation denoted by s^t. Specifically, at time t, the GRU cell takes in both the current embedding $\mathbf{F}_O^t$ and the previous cell state h^{t-1} to produce s^t and h^t, which can be formulated as

$$s^t, h^t = GRU(\mathbf{F}_O^{t-1}, h^{t-1}). \tag{5}$$

The state vectors, denoted as s^1,...,s^t, are processed through an actor-critic network for two primary purposes: firstly, to predict the conditioned action probability distribution $\pi_{\theta_1}(a^t|s^t)$ and secondly, to estimate the state value $V_{\theta_2}(s^t)$. The actor and the critic are implemented with a single linear layer parameterized by θ_1 and θ_2, respectively. The action sampler shown in Fig.1 samples the actual action denoted a^t, based on $\pi_{\theta_1}(a^t|s^t))$. The training aims to maximize the expected discounted return R:

$$R = \mathbb{E}_\pi[\sum_{t=1}^{T} \Gamma^t r(s^{t-1}, a^t)], \tag{6}$$

where Γ represents the discount factor, while T is the maximum number of time steps. Additionally, π denotes the policy that governs the robot agent. $r(s^{t-1}, a^t)$ is the reward given by the environment at the time step t. The reward is detailed in Section 3.3. Proximal Policy Optimization (PPO) [31] is adopted in this work to optimize Equation (6). The entire procedure is described in Algorithm 1.

3.3 Reward and Training

To address the backtracking issue discussed in Section 1, besides integrating the audio feature and the visual feature to guide the agent, as mentioned in Section 3.2, we implement a combination of fundamental rewards that facilitate navigation. These include rewards for stopping at the designated goal, time-related penalties, rewards for decreasing geodesic distance, and penalties for increasing geodesic distance. Additionally, we introduce specialized rewards, termed the directional continuity reward and historical trajectory reward.

Algorithm 1 Audio Visual Guided Navigation.

Require: Environment E, stochastic policies π, initial actor-critic weights θ_0, initial encoder and DSFI weights $\mathbf{W}_0$, the number of updates M, the number of episodes N, max time steps T.

Ensure: Trained weights: θ_M and $\mathbf{W}_M$

for $i = 1$ to M **do**
  # Run policy $\pi_{\theta_{i-1}}$ in environment for N episodes and T timesteps
  $\{(o^t, h^{t-1}, a^t, r^t)_i\}_{t=1}^{T} \leftarrow \text{roll}(E, \pi_\theta, T)$ for i-th update
  Compute advantage estimates
  # Optimize w.r.t. θ and $\mathbf{W}$
  $(\theta_i, \mathbf{W}_i) \leftarrow$ update the weights using the PPO algorithm to maximize Equation (6)

Directional continuity reward penalizes the agent for abrupt changes in its direction of movement. It encourages the agent to move smoothly and continuously. Let θ_{prev} be the agent's previous movement direction and θ_{curr} be its current movement direction. Then the directional continuity reward r_{dcr} can be be defined as:

$$r_{dcr} = -\beta \times |\theta_{curr} - \theta_{prev}| \tag{7}$$

where β is a constant that determines the penalty's strength for changing direction. Backtracking often requires an agent to make a U-turn or change its direction drastically. By penalizing sharp directional changes, the agent is less likely to make such U-turns, thus reducing backtracking.

Historical trajectory reward is based on storing the recent positions the agent has visited. If the agent revisits a location that has been in the recent past, it receives a penalty. Let H_i^t be the set of positions visited by the agent in the past i timesteps, and p_{curr} be the agent's current position. Then the historical trajectory reward r_{htr} at time t can be defined as:

$$r_{htr} = \begin{cases} -\alpha, & \text{if } p_{curr} \in H_i^t, \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where α is a positive constant representing the penalty for revisiting a recent position. By penalizing the agent for revisiting recent locations, the agent is discouraged from taking routes that lead it back to where it has just come from, thus reducing the chances of backtracking.

In our experimental framework, we amalgamate the aforementioned rewards with three straightforward incentives, which follow SoundSpaces [6]: (1) a reward of +10 points when the robot successfully navigates to the target and initiates the "Stop" action; (2) a reward of +0.25 points whenever the Manhattan distance

Table 1: **Quantitative comparisons with navigation methods.** We compare our model with existing methods on the Replica dataset and the Matterport3D dataset under both Heard and Unheard settings. SPL, SR, and SNA denote the success rate, success weighted by path length, and success weighted by number of actions, respectively.

Method	Replica						Matterport3D					
	Heard			Unheard			Heard			Unheard		
	SPL	SR	SNA	SPL	SR	SNA	SPL	SR	SNA	SPL	SR	SNA
RandomAgent	4.9	18.5	1.8	4.9	18.5	1.8	2.1	9.1	0.8	2.1	9.1	0.8
Direction Follower	54.7	72.0	41.1	11.1	17.2	8.4	32.3	41.2	23.8	13.9	18.0	10.7
Frontier Waypoints	44.0	63.9	35.2	6.5	14.8	5.1	30.6	42.8	22.2	10.9	16.4	8.1
Supervised Waypoints	59.1	88.1	48.5	14.1	43.1	10.1	21.0	36.2	16.2	4.1	8.8	2.9
LLA [8]	57.6	83.1	47.9	7.5	15.7	5.7	22.8	37.9	17.1	5.0	10.2	3.6
SoundSpaces [6]	78.2	94.5	52.7	34.7	50.9	16.7	55.1	71.3	32.6	25.9	40.1	12.8
AV-WaN [11]	86.6	**98.7**	70.7	34.7	52.8	27.1	72.3	**93.6**	54.8	40.9	56.7	30.6
AVGN (Ours)	**88.5**	97.3	**71.9**	**34.9**	**59.1**	**29.0**	**84.3**	88.3	**57.1**	**53.2**	**62.2**	**38.6**

between the robot and the target decreases; and (3) a time penalty of -0.01 for each action taken, promoting more efficient navigation.

4 Experiemnts

4.1 Dataset and Benchmark

In this study, we employ our method in conjunction with the 3D environment benchmark, SoundSpaces [6], [7]. SoundSpaces stands out as an advanced and lifelike acoustic simulation platform tailored for audio-visual embodied AI research. This platform is built based on Habitat [32], [33], offering a wide spectrum of research possibilities, ranging from audio-visual navigation and exploration to echolocation, as well as audio-visual floor plan reconstruction. This broad array of capabilities enables researchers to delve deeper into the nuances of embodied vision in various dimensions. For our experiments and analysis, we have specifically chosen datasets from Matterport3D [34] and Replica [35]. We use the AudioGoal setting for our experiments. We mainly use the telephone as a sound source for the experiments.

For evaluation, we apply three metrics in our experiments: 1) success rate (SR), the fraction of successful episodes, i.e., episodes in which the agent stops exactly at the audio goal location on the grid; 2) success weighted by path length (SPL) [36], the standard metric that weighs successes by their adherence to the shortest path; 3) success weighted by number of actions (SNA), which penalizes rotation in place actions, which do not lead to path changes.

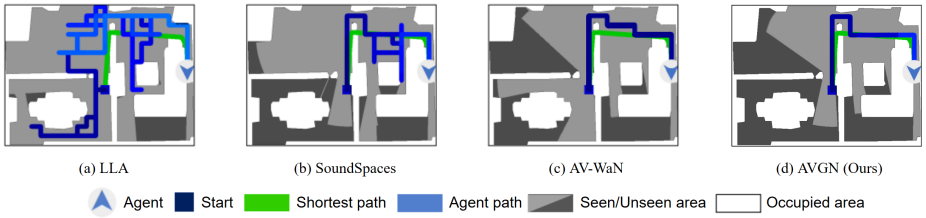

Fig. 3: **Comparison with state-of-the-art methods.** Our results showcase smooth navigation without backtracking problems.

4.2 Implementation Details

In the implementation of the AVGN model, raw egocentric vision images of size $256 \times 256 \times 3$ are processed through 6 transformer blocks, each with multi-head self-attention mechanisms, producing a visual feature map of $64 \times 64 \times 128$. Currently, the audio data, represented as spectrograms with dimensions 64×128, undergoes processing through a CNN composed of three convolutional layers (each followed by batch normalization and ReLU activation), resulting in a feature map of $8 \times 16 \times 64$. These visual and audio features are fused using the Dual Stage Feature Integration (DSFI) mechanism, yielding a combined embedding of dimension 128. This embedding undergoes temporal refinement in a bidirectional GRU with a hidden state of 512, feeding into the actor-critic framework. The actor and critic networks, each built from two dense layers with 256 neurons and ReLU activations, optimize actions using the Proximal Policy Optimization (PPO) algorithm [31] at a learning rate of 1×10^{-4}.

4.3 Comparison with Baselines

The experiment results are shown in Table 1. We compare our methods with baselines on two datasets: Replica and Matterport3D. On each dataset, we evaluated the methods under two settings: Heard and Unheard, where Heard stands for the standard AudioGoal setting with both visual and audio sensors equipped on the agent, while in the Unheard setting, only the visual sensor is attached. The existing methods and baselines with which we compared are as follows:

1. **Random Agent**: An agent that randomly selects each action and selects *Stop* when it reaches the goal.
2. **Direction Follower**: A hierarchical model that sets intermediate goals K meters in the predicted direction of arrival (DoA) of the audio. Following [11], we set $K = 2$ in Replica and $K = 4$ in Matterport3D.
3. **Frontier Waypoints**: A hierarchical approach that combines the predicted DoA with the boundaries of the explored areas to designate the next waypoint.
4. **Supervised Waypoints**: This model employs supervised learning to predict waypoints within its field of view (FoV), using RGB frames and audio spectrograms.

5. **SoundSpaces** [6]: This model applies CNNs to both acoustic and visual inputs from sensors, using a simple concatenation to combine different modalities. The output is passed to a GRU, and an actor-critic model estimates the value of the state and policy distribution.
6. **LLA** [8]: A conventional phase-based navigation method that uses a topological graph for enhanced path planning.
7. **AV-WaN** [11]: This method utilizes waypoints, represented through a topological graph, to improve navigation efficiency over longer distances. Similar to SoundSpaces, the model uses a simple concatenation to combine different modalities.

In the Replica dataset, under the Heard setting, the proposed AVGN method conspicuously stands out. It achieves an SPL score of 88.5, suggesting optimal path utilization. While its success rate of 97.3 slightly lags behind the state-of-the-art method AV-WaN (98.7), AVGN leads the SNA metric with a dominant score of 71.9, a roughly 1.2 point improvement. In the Unheard setting, AVGN achieves the best performance, with 34.9 in SPL, 59.1 in SR, and 29.0 in SNA. Transitioning to the Matterport3D dataset, under Heard conditions, the performance of AVGN is unparalleled. The SPL, SR, and SNA are 84.3, 88.3, and 57.1 respectively. This places AVGN firmly at the top, especially in the SPL metric, where it outperforms the nearest competitor by a substantial 12 points. Under the Unheard setting, AVGN's excellence persists. It registers the highest scores across all three metrics: 53.2 in SPL, 62.2 in SR, and 38.6 in SNA. To put this into perspective, in the SPL metric alone, AVGN exceeds the closest rival by an impressive 12.3 points.

When the performance of AVGN is compared with LLA, SoundSpaces and AV-WaN, the progress is evident. Particularly in the Matterport3D dataset under the Heard setting, the performance improvements of AVGN are significant. It not only tops the charts, but does so with a clear margin in the SPL, highlighting the advancements made from baseline methodologies to our proposed methods. In essence, the AVGN method introduced in this study signifies a landmark in embodied AI navigation research. Its consistent top-tier performance, especially when evaluated against the backdrop of prior methods, underscores its potential and the considerable progress made in the field.

We also include qualitative results for our model and comparisons with LLA [8], SoundSpaces [6] and AV-WaN [11] in Fig.3. These qualitative results intuitively highlight the strengths and advantages of our methods.

4.4 Ablation Study

In our research, we conduct an ablation study using three distinct settings: (1) without the DSFI module and backtracking rewards; (2) without backtracking rewards; and (3) without the DSFI module. To evaluate the efficacy of our proposed model, we compare these settings within the Heard configurations across both datasets. Detailed outcomes of these evaluations are presented in Table 2. We denote the Directional continuity reward and Historical trajectory reward as R_b for simplicity.

Table 2: **Ablation study for AVGN.** We evaluate our model under three configurations: 1. AVGN without both DSFI and backtracking rewards; 2. AVGN excluding backtracking rewards; and 3. AVGN without the DSFI module.

Method	Replica			Matterport3D		
	SPL	SR	SNA	SPL	SR	SNA
AVGN w/o DSFI and R_b	74.9	88.5	51.8	54.9	78.5	31.8
AVGN w/o R_b	84.7	92.0	61.1	81.1	87.2	48.4
AVGN w/o DSFI	74.0	83.9	55.2	56.5	74.8	35.1
AVGN (Ours)	**88.5**	**97.3**	**71.9**	**84.3**	**88.3**	**57.1**

Table 2 elucidates the pivotal roles of the Dual Stage Feature Integration (DSFI) mechanism and the backtracking rewards in our AVGN model through an ablation study. By assessing the model's performance across different configurations on the Replica Heard and Matterport3D Heard datasets, we draw several conclusions as follows. When both components are removed from the model (AVGN w/o DSFI and R_b), there is a pronounced decline in performance across all metrics. The SPL values for Replica and Matterport3D are notably reduced to 74.9 and 54.9, respectively, indicating that both the DSFI and backtracking rewards are key contributors to the model's efficacy. By retaining only the DSFI and omitting backtracking rewards (AVGN w/o R_b), the model achieves an SPL of 84.7 on the Replica dataset. This substantial improvement from the first configuration underlines the DSFI's indispensable role in synergistically fusing visual and audio inputs. The version without the DSFI but including backtracking rewards (AVGN w/o DSFI) performs similarly to the model lacking both components. However, when comparing the full AVGN with the model that only has DSFI, we see better results with the complete AVGN. This suggests that backtracking rewards contribute positively to the DSFI mechanism. Our fully-fledged AVGN model, which integrates both the DSFI and backtracking rewards, unsurprisingly registers the highest performance metrics on both datasets. With outstanding SPL scores of 88.5 and 84.3 for Replica and Matterport3D, it reiterates the combined strength of both the DSFI and the backtracking rewards. Collectively, this ablation study reaffirms the centrality of the DSFI in our model. Its prowess in amalgamating visual and audio information proves vital in enhancing the robot's navigation capabilities.

5 Conclusions

In this work, we introduced the Audio-Visual Guided Navigation (AVGN) approach, a novel audio-visual navigation methodology. Central to AVGN is the Dual Stage Feature Integration (DSFI), an advanced feature fusion module adept at amalgamating audio and visual cues. Furthermore, we incorporated a triad of rewards specifically designed to address the backtracking issue. Comparative evaluations have demonstrated the superior performance of AVGN over exist-

ing methods. As a future direction, we aim to explore audio-visual navigation challenges in more complex scenarios, such as environments with moving sound sources or those with background noise.

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